

Distribution function list (under construction)

Processing target of
distribution calculation

Distribution Generation

bunpu_data()
vector_gene (def)
bunpu_gene (def)
bunpu_hist()

Distribution calculation

bunpu_add (def)
bunpu_sub (def)
bunpu_product (def)
bunpu_division (def)
bunpu_multiple (def)

Distribution Graph

bunpu_graph (def)
bunpu_simu_graph (def)
bunpu_twin_graph (def)
bunpu_trace_graph (def)
bunpu_mean (def)
bunpu_sdev (def)
bunpu_weibull (def)
bunpu_normal (def)
bunpu_percent (def)
bunpu_percent2 (def)
bunpu_percent3 (def)
bunpu_balance()
bunpu_ruiseki (def)
bunpu_ruiseki2 (def)

Distribution and other functions

bunpu_mean (def)
bunpu_sdev (def)
bunpu_weibull (def)
bunpu_normal (def)
bunpu_file()
bunpu_save()
bunpu_load()

Processing target of
distribution integration

Integral Function

bunpu_simu_start (def)
bunpu_simu_integral (def)
bunpu_simu_readout (def)

Calculations in the integration loop

bunpu_simu_add (def)
bunpu_simu_sub (def)
bunpu_simu_invsb (def)
bunpu_simu_prd (def)
bunpu_simu_div (def)
bunpu_simu_invdiv (def)
bunpu_simu_calc (def)
bunpu_simu_arctan (def)
bunpu_simu_tan (def)
bunpu_simu_sin (def)
bunpu_simu_arcsin (def)
bunpu_simu_cos (def)
bunpu_simu_arccos (def)
bunpu_simu_dist (def)
bunpu_simu_comp (def)
bunpu_simu_comp_prd (def)
bunpu_simu_prb (def)
bunpu_simu_limit2 (def)
bunpu_simu_limit (def)
bunpu_simu_limitspd (def)
bunpu_simu_inprod (def)
bunpu_simu_outprod (def)
bunpu_simu_ones (def)
bunpu_simu_zeros (def)
bunpu_simu_mean (def)
linemap_shape (def)
linemap_extrct (def)
linemap_input (def)

Note

The bunpu function has different internal processing targets for distribution calculation and distribution integration.
Distribution information is stored in the following class variables.

Processing target of distribution calculation
Information that quantifies the distribution profile
bunpu.para,bunpu.mesh,bunpu.flatten

Processing target of distribution integration
Includes initial information and post-change information (time series information, etc.) required for integration
bunpu.linemap

Mutual conversion between calculation and integration
bunpu_start: generates linemap from para,mesh
bunpu_simu_readout: generates para,mesh from linemap

List of distribution graph functions

bunpu1.bunpu_graph (filename=text1,{gr=para1}, {levels=para2}, {linewidth=para3}, {yscale=para4}, {xscale=para5}, {gpu=para6}, {ruiseki=para7}, {view = para8}, {fs = para9}))

Generate a graph of the distribution

bunpu1:The instance of the distribution for which you want graph

text1:The file name of the graph to generate

para1:If not specified or set to 0, a 3D graph will be displayed. If set to 1, a contour graph will be displayed.

para2:Number of contour lines on a 2D graph

para3:Line thickness for 1D graphs

para4:Specify the y-axis range

para5:Specify the x-axis range

para6:Acceleration with GPU computing(requires cuda and cupy)

para7:Show cumulative distribution

para8:In the case of 3D, change the perspective of the graph

para9:Font size of characters(If not specified, it is set to 14)

bunpu1.bunpu_percent (filename=array1,pos1=array2,pos2=array3, dirc=array4, {shw = para1}, {view = para2}, {fs = para3}))

Calculate percentiles and generate graphs to display them

If pos1 and pos2 are the same array, calculate the area (probability) in the dirc direction. If they are different, calculate the area between or outside.

If it is two or more dimensions, calculate the above by the line or surface that passes through pos1 and pos2 and is perpendicular to dirc.

bunpu1:The instance of the distribution for which you want the percentile

array1:The file name and parameter name array for the graph to be generated

array2:Boundary position 1

array3:Boundary position 2

array4:Percentile direction, or boundary normal if in 2 or more dimensions

para1:if 1 graph is displayed, if 0 the percentile probability Values is output

para2:In the case of 3D, change the perspective of the graph

para3:Font size of characters(If not specified, it is set to 14)

bunpu1.bunpu_percent2 (bunpu2,filename=array1,dirc=array2, {shw = para1}, {view = para2}, {fs = para3}))

Comparison of parameter values of two distributions

Compare distributions, calculate the probability of exceeding a specified direction, and generate a graph to display the results

bunpu1:Instances of distributions to compare as probability distributions

bunpu2:Instances of distributions to compare as cumulative distributions

array1:The file name and parameter name array for the graph to be generated

array2:Cumulation direction of cumulative distribution

para1:if 1 graph is displayed, if 0 the area of the achievement distribution is output

para2:In the case of 3D, change the perspective of the graph

para3:Font size of characters(If not specified, it is set to 14)

List of distribution graph functions

bunpu1.bunpu_percent3 (filename=array1,pos=array2, dirc=array3, , {shw = para1}, {view = para2}, {fs = para3})

Calculates the probability of the range enclosed by multiple boundaries specified in a 2D distribution, and generates a graph to display it.

You can specify multiple boundaries as many as the number of elements in the pos array.

3D is not yet implemented.

bunpu1:The instance of the distribution for which you want the percentile

array1:The file name and parameter name array for the graph to be generated.

array2:Specify the border position [[x1,y1],[x2,y2],[x3,y3]]

array3:Specify the inclusive direction of the border [[x1,y1],[x2,y2],[x3,y3]]

para1:if 1 graph is displayed, if 0 the area of the achievement distribution is output

para2:Change the perspective of the graph if 3rd dimension is implemented

para3:Font size of characters(If not specified, it is set to 14)

Function arguments

para:numerical value

array:array

text:text

textarray:text array

bunpu:Distribution Instance

{ }:If the contents of these brackets are not specified, the default value will be used (elements without brackets are required)

bunpu1.bunpu_balance(bunpu2, filename=array1, {dirc=array2}, {shw = para1} , {view = para2}, {fs = para3})

Intersection of cumulative distributions(It is assumed that bunpu.volume has a value, currently only one dimension)

Search for parameter values where two distributions intersect

The balance point between supply and demand, etc.

Implemented only in 1D, not implemented in 2D or 3D

bunpu1:The first distribution instance to balance

bunpu2:The second distribution instance that balances

array1:The file name and parameter name array for the graph to be generated.

array2:bunpu1とbunpu2の累積方向、未指定でbunpu1がマイナス方向bunpu2がプラス方向

para1:if 1 graph is displayed, if 0 the area of the achievement distribution is output

para2:Change the perspective of the graph if 3rd dimension is implemented

para3:Font size of characters(If not specified, it is set to 14)

bunpu1.bunpu_twin_graph (bunpu2 , filename=text1 , contact=para1, {view = para2}, {fs=para3})

Probability of approach between probability distributions

Gives the probability that two distributions are within a given distance

bunpu1:The first distribution instance to evaluate the distance

bunpu2:The second distribution instance to evaluate the distance

text1:The file name of the graph to generate.

para1:Specify the distance to be considered as approaching

para2:In the case of 3D, change the perspective of the graph

para3:Font size of characters(If not specified, it is set to 14)